RESIDUAL LIFE ASSESSMENT OF ALUMINA BASED CERAMIC COATINGS

A Project Report Submitted
In Partial Fulfillment of the Requirements
for the Degree of

BACHELOR OF TECHNOLOGY
in
MECHANICAL ENGINEERING

by

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SHAKTI KUMAR (1873440054)
SURAJ KUMAR (1873440059)
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to the
Faculty of Mechanical Engineering Department

DR. A. P. J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
May, 2022

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CERTIFICATE

Certified that **Ram Prasad (Roll No.-**1873440044), **Shakti Kumar (Roll No.-**1873440054), **Suraj Kumar (Roll No.-**1873440059), **Jitendra Kumar Chaurasiya (Roll No.-**1873440024), **Prince Kumar (Roll No.-**1873440040) have carried out the research work presented in this project entitled **"Residual Life assessment of Alumina Based Ceramic Coatings"** for the award of **Bachelor of Technology** from Dr. A. P. J. Abdul Kalam Technical University, Lucknow, under my supervision. The project embodies the results of original work, and studies are carried out by the student their self and the contents of the project do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University / Institution.

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ABSTRACT

Non-destructive and accurate measurement of residual stress in ceramic coatings is challenging, but it is crucial to the assessment of coatings' failure and life. In this project, the thermal-cycle-dependent residual stress in the samples of an atmosphere plasma sprayed thermal barrier coating systems has been non-destructively and accurately measured using optical spectroscopy which were thermally cycled at different high temperatures and over many cycling periods.

ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the B. Tech Project undertaken during B. Tech. Final Year. We owe special debt of gratitude to Designation, **Mr. Saurabh Tripathi**, Head of Department of Mechanical Engineering, and **Dr. Ashutosh Tiwari**, Department of Applied Science and Humanities, Rajkiya Engineering College, Banda for his constant support and guidance throughout the course of our work. His sincerity, thoroughness and perseverance have been a constant source of inspiration for us. It is only his cognizant efforts that our endeavors have seen light of the day.

We also take the opportunity to acknowledge the contribution of **Mr. Deep Singh Thakur**, Department of Mechanical Engineering, Rajkiya Engineering College, Banda, for his full support and assistance during the development of the project.

We also do not like to miss the opportunity to acknowledge the contribution of all faculty members of the department for their kind assistance and cooperation during the development of our project. Last but not the least; we acknowledge our family & friends for their contribution in the completion of the project.

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UNDERTAKING

We **Ram Prasad (Roll No.-**1873440044), **Shakti Kumar (Roll No.-**1873440054), **Suraj Kumar (Roll No.-**1873440059), **Jitendra Kumar Chaurasiya (Roll No.-**1873440024), **Prince Kumar (Roll No.-**1873440040)Student of Bachelor of Technology, **Mechanical Engineering, Rajkiya Engineering College, Banda,** hereby declare that the work presented in this report is outcome of our work correct to the best of our knowledge and this work presented does not infringe any patented work and has not been submitted to any university for the award of any degree or any professional use. Any work presented as a reference in this report has been cited properly. If it is found incorrect then its responsibility is on us.

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CHAPTER 1

INTRODUCTION

1.1 Thermal Barrier Coatings (TBC)

Thermal barrier coatings are insulating ceramic coatings deposited on bond coated superalloys, and are accustomed lower the substrate temperature and consequently improve element sturdiness and increase engine efficiency and potency in aircraft and industrial gas turbines. thermal barrier coatings (TBCs) modify the engines to operate at higher temperatures whereas not raising the lower part of metal temperatures victimization cooling systems among the recent section components so, enhance the in operation potency of the engines [1]. Therefore, continuing development of TBCs is important to extend the inlet gas temperature additional for rising the performance of gas turbines. Hence, TBCs with low thermal conductivity, state and phase stability, and high resistance to sintering have ever increasing demands.

Thermal barrier coatings for insulating elements in heat engines are underneath development. Advances during this field are accelerated through analysis to develop coatings for air-cooled turbine blades and vanes. the advantages of thermal hairier coatings includes lower metal temperatures, moderation of thermal transients, reduced needs for cooling air and/or higher gas temperatures. The result is improved performance and potency, longer element lifetimes, simplified designs and, in certain cases, the substitution of less costly bimetallic substrates.

TBCs can build the cooling of the underlying substrate, leading to improved partly strength. Meanwhile, fuel efficiency are inflated by allowing an increase inside the recess temperature in a} very rotary engine. Commonly, a TBC consists of a ceramic top coat, a bond coat, and a thermally grownup oxide (TGO) layer forming at operation temperatures. concerning the function and sturdiness of TBCs, the performances of thermal insulation and resistance to spalling throughout high-temperature services are crucial [2].

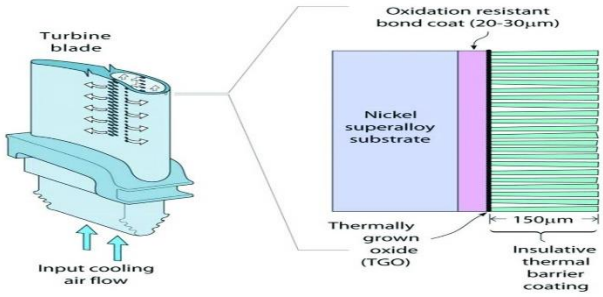


Fig. 1.1 Structure of Thermal Barrier Coatings[3]

1.2 Structure

Generally, thermal barrier coatings usually comprises four layers: the metal substrate, metallic bond coat, thermally-grown oxide compound (TGO), and ceramic topcoat. This ceramic layer creates the biggest thermal gradient of the TBC and keeps the lower layers at a lower temperature than the surface. A bonding layer is applied between the metal substrate and also the overcoat to shield the metal substrate from oxidation reaction and corrosion at high temperatures corrosion and assist the coupling of the ceramic top coat and the metal like substrate [4].

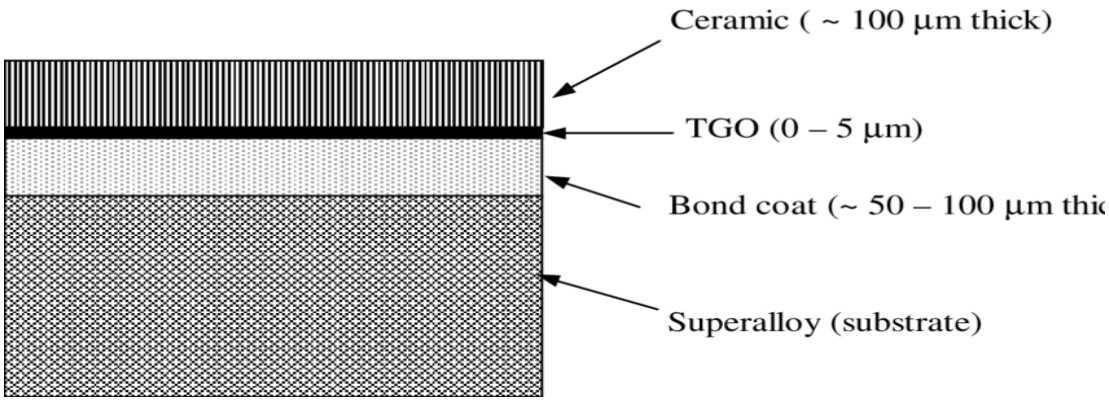


Fig. 1.1 : Thermal barrier coating structure[5]

1.3 Main Requirement for TBCs

TBCs should have low weight and low thermal conduction and that they ought to face up to massive stress variations because of heating and cooling, further as thermal shock. they have to be with chemicals compatible with the underlying metal and also the thermally adult chemical compound (TGO) and may operate in AN oxidizing setting. TBCs should give thermal insulation to the underlying superalloy engine components. they have to have strain compliance so as to reduce the thermal enlargement twin stresses with the superalloy components. in addition, they have to mirror abundant of the refulgent heat from the new gas and thereby, preventing it from reaching the superalloy substrate. Further, TBCs should give thermal protection to the substrate for prolonged service times and thermal cycles while not getting to the condition of failure.

1.4 Properties

A good thermal stability, an occasional thermal conduction, a high constant of thermal enlargement (CTE) together with a high fracture toughness are the most needed properties for the ceramic high coat on high of bimetal parts. The YSZ encompasses a high freezing point (2700ºC) and one of the lowest thermal conductivities of all

ceramics at elevated temperatures; the conduction of bulk YSZ and YSZ wholly totally different microstructures and consistence were reported to be a combine of.6 W/mK (5.3 wt.% YSZ, 600ºC) and 0.7-1.4 W/mK (7.25 wt.% YSZ), severally. The YSZ conjointly encompasses a high CTE (11 nine 10-6 K-1), that is about to that of the underlying superalloy substrate (14 nine 10-6 K-1) and helps to mitigate stresses arising from the thermal enlargement twin. however a twin still remains and these stresses result in crack propagation at intervals the coatings despite the high toughness of seven wt.% YSZ (phase composition and transitions are going to be careful below). Therefore, primarily by attempting to scale back the strain levels and/or increasing the strain tolerance of the coatings, an extra improvement of the coating performance is desired.

1.5 Application of TBCs

TBCs give thermal insulation to superalloy engine components like the combustor, rotating blades, stationary guide vanes, blade outer air-seals, shrouds within the aggressive section behind the combustor, and once burners within the tail section of jet engines. important gas-temperature increase is achieved by victimization TBCs in association with innovative air-cooling approaches than that obtained by earlier materials as well as single-crystal Ni-based superalloys [6]

1.6 Degradation Mechanism

During service, many kinetic processes occur in parallel. Interdiffusion happens between the bond coat and also the superalloy. Consequently, Al diffuses from the bond coat to create the TGO. Microstructural, chemical, and section changes occur altogether the materials including the ceramic high coat. The rates of those thermally activated processes are expected to extend exponentially with temperature. These processes typically result in degradation and failure of the coating [7]. throughout service, failure of the TBC system happens relying upon the subsequent 3 factors:

1.6.1 Bond Coat Degradation

Bond coat plays a vital role in promoting the sturdiness of the TBC system. however the role of the bond coat is incredibly advanced and poorly understood. In most sensible cases, oxidization of the bond coat becomes the predominant coating failure mechanism. throughout high temperature exposure, NiCrAlY bond coat is oxidised leading to a TGO layer on the bond coat. once reaching a essential thickness, the TGO becomes liable to nuclear reaction, that successively results to the failure of the TBC system.

1.6.2 Generation of High Residual Stress

Residual stress contains a vital impact on the performance of a TBC system. The role of residual stress is extremely complicated and varies with the distinction in system configurations. Thermal growth twin between the 3 layers generates residual stress leading to degradation of a TBC system. High residual stresses are elicited within the TBC because of thermal enlargement mismatch and bond coat (BC) oxidization resulting in failure by spalling and delamination. Raman spectroscopic analysis and the curvature strategies are used to measure residual stresses.

1.6.3 Top Coat Degradation

Top coat degradation is another parameter that governs TBC failure. The ceramic top coat contains a tendency to crack because of stress generated from thermal enlargement mismatch between the 3 layers of the TBC system. once the top coat cracks, O_2 gas simply diffuses to the bond coat resulting in the harmful failure of the TBC system.

1.7 Residual Life Assessment (RLA)

Equipment was designed to be worked under some conditions and has some expected life. If these equipments are operated under harsher conditions then their useful life is small. For the estimation of the residual lifespan the utmost expected time period, assumptive that the instrumentality are operated underneath same conditions, has got to be found out. The expected period of your time could also be a function of the operative conditions. In dependence of the common load level it'll get shorter for higher load conditions. The residual amount of your time could also be outlined as a result of the difference between the expected amount of your time and the actual age.

1.8 Process techniques for TBCs

In industry, thermal barrier coatings are produced in a number of ways:

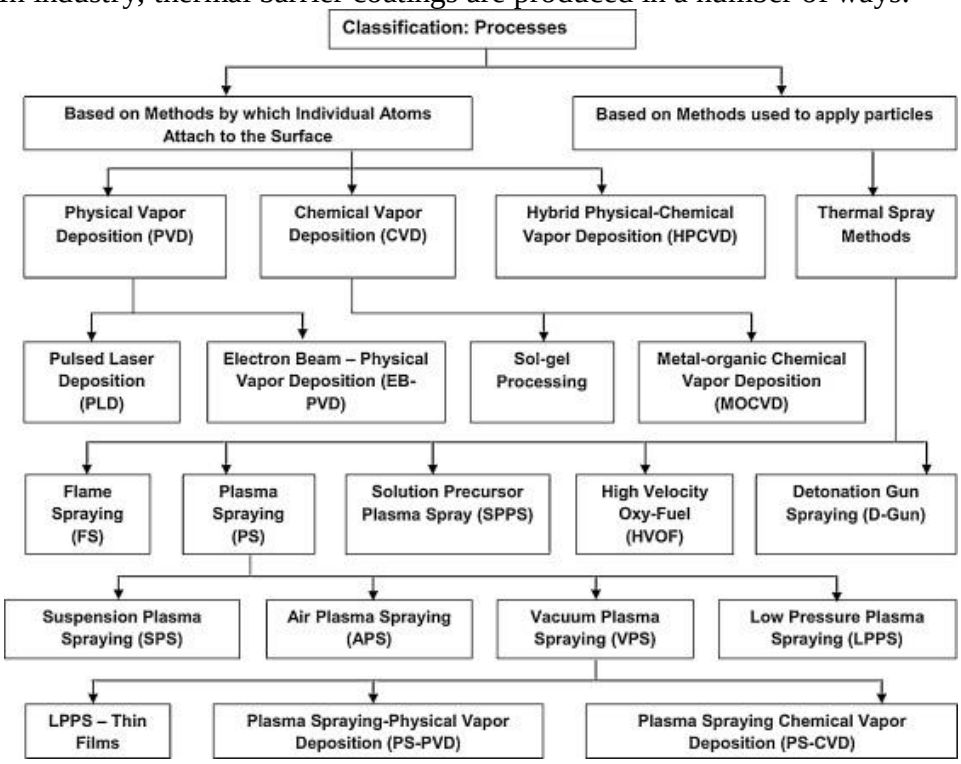


Fig. 1.8 Processing and design methodologies for advanced and novel thermal barrier coatings for engineering applications[8]

Among all these processes, Air Plasma Spraying (APS) method is used.

1.8.1 APS Process

In the APS method, ceramic powders are introduced into a hot temperature plasma plume, liquified within the plume, and accelerated towards the substrate whereby liquid droplets spread over and type splats that are speedily quenched. In one pass, many consecutive splats are deposited on the substrate and also the coating thickness is accumulated by suggests that of many passes. A typical broken cross-sectional of the plasma sprayed ceramic coating show layers of splats beside interlamellar pores, cracks, and spherical pores . Coating compliance is accumulated by the presence of the cracks and thereby, extending their lifetimes [9]. The ensuing coating microstructure is powerfully dependent on process conditions like spray parameters (e.g., torch current, plasma gas rate, carrier gas rate, torch traverse speed, and stand off distance) and feedstock materials (e.g., size, temperature, and velocity). Splat morphologies are modified with the angle of impact of happening particle . Higher substrate temperatures result in lower consistency and improved inter-splat contact leading to increased coating properties [10]. throughout service operations at high temperatures, a TGO layer, primarily an Al_2O_3 layer, is developed between the bond coat and also the high coat thanks to the oxidization of the bond coat. this can be the foremost vital issue that

determines the life of the TBC system. The thickness of this layer will increase with increasing operation time. High stresses are present at the bond coat and TGO interface due to compound layer growth, thermal enlargement anomaly, and applied hundreds. As a result, crack initiates and propagates leading to spallation of the ceramic layer, and eventually, system degradation. throughout thermal exposure at ≥1,000°C, Ni(Cr, $Al)_2O_4$ (spinels) and NiO clusters are shaped at the interface of the Al_2O_3 layer and also the ceramic coating within the TBC system with MCrAlY (M=Ni, Co) bond coat. Cracks were nucleate on these compound clusters and grew into the ceramic coating resulting in premature TBC separation.

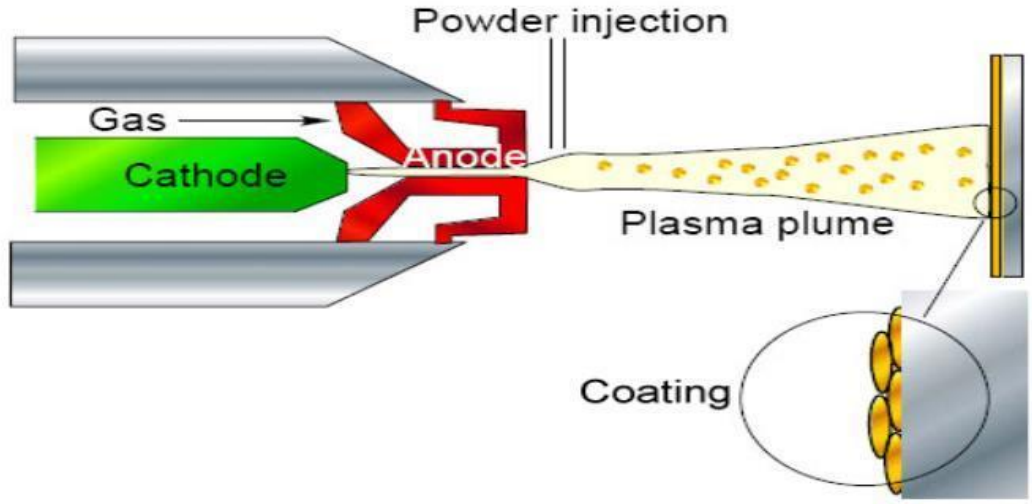


Fig. 1.8.1 APS Process[11]

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Survey

Paper [1] Thermal Shock and cycling Behavior of thermal barrier Coatings (TBCs).

Abdullah Cahit Karaoglanli, Kazuhiro Ogawa,Ahmet Türk and Ismail Ozdemir, Thermal Shock and cycling Behavior of thermal barrier Coatings (TBCs) utilized in employed in [12] has presented turbine engines work as an influencing power generating facility and are utilized in aviation trade to provide the thrust by changing combustion product into mechanical kinetic energy. Basic issues relating to the enhancements in trendy turbine engines are higher potency and performance. Increase in power and potency of rotary engine| turbine engines are often achieved through increase in turbine recess temperatures. The materials used ought to have excellent mechanical strength and corrosion resistance and therefore be ready to work beneath aggressive environments and high temperatures. The temperatures that rotary engine blades are exposed to can be near to the melting point temperature of the super alloys. Internal cooling by cooling channels and insulation by thermal barrier coatings (TBCs) is employed so as to lower the temperature of rotary engine blades and stop the failure of super alloy substrates.

Paper [2] Design and optimization of coating structure for the thermal barrier coatings fabricated by atmospheric plasma spraying via finite element method.

L. Wang, X.H. Zhong, Y.X. Zhao, S.Y. Tao, W. Zhang, Y. Wang, X.G. Sun, ―Design and optimization of coating structure for the thermal barrier coatings fabricated by atmospheric plasma spraying via finite element method [12] has presented fabricating the thermal barrier coatings (TBCs) with excellent performance is to find an optimized coating structure with high thermal insulation effect and low residual stress. This paper discusses the design and optimization of a suitable coating structure for the TBCs prepared by atmospheric plasma spraying (APS) using the finite element method. The design and optimization processes comply with the rules step by step, as the structure develops from a simple to a complex one. The research results indicate that the suitable thicknesses of the bond-coating and top-coating are 60–120 μm and 300–420 μm, respectively, for the single ceramic layer YSZ/NiCoCrAlY APS-TBC.

Paper [3] Selective coatings for new ideas of parabolic trough collectors.

J. Barriga, U. Ruiz-de-Gopegui, J. Goikoetxeaa, B. Coto, H. Cachafeiro, ―Selective coatings for new ideas of parabolic trough collectors [13] has given the CSP technology

supported parabolic trough reflector for giant electricity generation functions is presently the foremost mature of all CSP designs and styles in terms of previous operation expertise and scientific and technical analysis and development. the present parabolic trough style deals with a most operational temperature around 400ºC within the absorbent collector tube however some recent designs and styles are planned to extend the operating temperature to 600ºC increasing the performance by 5-10% to achieve the improved productivity that the market demands. These systems are expected to be operating throughout 20-25 years. one in every of the key points of the receiver is that the stack of layers forming the selective absorbent coating. With this new style the coating needs to fulfill new needs because the collector are operating at600ºC and during a low vacuum of 10-2 mbar.

Paper [4] Experimental Study of Ceramic Coated Piston Crown for Compressed Natural Gas Direct Injection Engines.

Helmisyah Ahmad Jalaludin, Shahrir Abdullah, Mariyam Jameelah Ghazali, Bulan Abdullah, Nik Rosli Abdullah,―Experimental Study of Ceramic Coated Piston Crown for Compressed Natural Gas Direct Injection Engines [14] has presents High temperature produced in a compressed natural gas with direct injection system (CNGDI) engine may contribute to high thermal stresses. Without appropriate heat transfer mechanism, the piston crown would operate ineffectively. Bonding layer NiCrAl and ceramic based yttria partially stabilized zirconia (YPSZ) were plasma sprayed onto AC8A aluminium alloy CNGDI piston crowns and normal CamPro piston crowns in order to minimize thermal stresses.

Paper [5] Characterisation of Thermal Barrier Coating on Piston Crown for Compressed Natural Gas Direct Injection (CNGDI) Engines.

Helmisyah A.J., Ghazali M.J., ―Characterisation of Thermal Barrier Coating on Piston Crown for Compressed Natural Gas Direct Injection (CNGDI) Engines [13] has presented the high temperature and pressure produced in an engine that uses compressed natural gas with direct injection system (CNGDI) may lead to high thermal stresses. The piston crown fails to operate effectively with insufficient heat transfer. In this study, partially stabilized zirconia (PSZ) ceramic thermal barrier coatings were plasma sprayed on CNGDI piston crowns (AC8A aluminium alloys) to reduce thermal stresses. Several samples were deposited with NiCrAl bonding layers prior to the coating of PSZ for comparison purposes. Detailed analyses of microstructure, hardness, surface roughness, and interface bonding on the deposited coating were conducted to ensure its quality. High stresses were mainly concentrated above the pinhole and edge areas of the piston. In short, the PSZ/ NiCrAl coated alloys demonstrated lesser thermal stresses than the uncoated piston crowns despite a rough surface. Extra protection is thus given during combustion operation.

Paper [6] Coating thickness and roughness impact on stress distribution of A356.0 beneath thermomechanical loadings.

A. Moridi, M. Azadi and G.H. Farrahi, ―Coating thickness and roughness impact on stress distribution of A356.0 beneath thermomechanical loadings [15] has given forged aluminium-silicon alloy, A356.0, is wide utilized in automotive elements like ICE cylinder heads and additionally in part industries due to its outstanding mechanical, physical, and casting properties. boundary coatings are applied to combustion chamber so as to scale back fuel consumption and pollutions and additionally improve fatigue lifetime of elements. However, studies on behaviour of A356.0 with thermal barrier coating are still rare. the aim of the current work is to simulate stress distribution of A356.0 under thermo-mechanical cyclic loadings, employing a two-layer elastic-viscoplastic model of ABAQUS code. The results of stress strain physical phenomenon loop are valid by associate degree out of section thermo-mechanical fatigue check. Ceramic coating thickness impact on stress distribution of take a look at specimens is investigated. completely different thicknesses from three hundred to 800 microns of prime coat and additionally roughness of the interfaces are simulated to induce best stress gradient which might cause an improvement of fatigue life.

Paper [7] A new high temperature resistant glass–ceramic coating for gas turbine engine components

SOMESWAR DATTA* and SUMANA DAS- A newer extreme temperature resistant glass–ceramic coating for turbine engine [16] parts has conferred that, the coatings exhibit high thermal shock resistance (no splintering or spalling) on ten cycles of thermal shock between temperature and also the operating temperature. any they found that ZnO– Al_2O_3–SiO_2 primarily based coating (ARDB-3) exhibits negligible increase in weight once one hundred h of static oxidization test at 1000–1050°C. the method of application of the developed coating materials to yield extreme temperature resistant coatings on nickel base super alloys for turbine application has been standardized. The ZnO– Al_2O_3–SiO_2 primarily based coating (ARDB-3) is appropriate for continuous use at 1000°C temperature. The developed coatings exhibit high abrasion resistance (loss in weight worth of 0×4–0×6 mg/cm2 in fifty,000 cycles in P.E.I. abrasion tester). Finally they found 2 new ceramic compositions (based on magnesium– aluminium–titanate and ZnO– Al_2O_3–SiO_2 glass systems) appropriate for application on nickel base super alloy.

Paper [8] Ceramic Coatings for High-Temperature Protection of Steel.

William N. Harrison, Dwight G. Moore, and Joseph C. Richmond _Ceramic Coatings for High-Temperature Protection of Steel [17] have found a new type of ceramic during war at the National Bureau of Standards and was used by the Army and Navy on the exhaust systems of certain aircraft and other vehicles. Laboratory and service tests have

shown these new coatings to be superior to conventional porcelain enamels for high-temperature service. The outstanding feature of the coatings are (a) high resistance to chipping under repeated severe thermal shock, (b) protection of the metal against oxidation during prolonged exposure in air at temperatures about 1,250° F, (c) freedom from the cracking and blistering produced in conventional enamels under comparable conditions of high temperatures and severe thermal gradients.

Paper [9] Failure processes within ceramic coatings at high temperatures.

Christopher C. Berndt - Failure processes within ceramic coatings at high temperatures [18] has used the methodology of Acoustic emission to study the failure processes of plasma sprayed NiCrA1Y + ZrO_2-12wt % Y_2O_3 coatings. Several techniques of analysis have been used to describe the mechanism of failure in relative terms of the number of cracks and their activity, i.e., their initiation or growth. These studies have focussed on examining the cracking behaviour of plasma-sprayed coatings by acoustic emission methods. Two specimens, identical except for the ceramic coating thickness, have been studied by repeated thermal cycling to 1150 ° C.

On the basis of these studies and results we are interested in the "Residual life assessment of Alumina based Ceramic Coatings".

2.2 Research Gap

After reading various research papers and going through different books and articles about Ceramic Coatings for UHTC applications, following conclusions are made:-

- Al_2O_3 used as coatings on the surface is very brittle in nature due to which it gets divided into too many small parts when it is subjected into high thermal stresses.
- So, to prevent this cause Yttria-stabilized Zirconia (YSZ) is added to provide mechanical stability.
- However in the environment of ultra high temperature due to the generation of high thermal stresses the coatings rupture and cracks generated and have tendency to propagate which is undesirable phenomena.
- So, to reduce this propagation of cracks, Carbon Nano Tubes (CNTs) are used in the Coatings which obstruct the propagation of cracks from both the sides.

2.3 Problem Formulation

- All these problems encountered is need to be addressed which leads and motivated us to carry out this Research based Project for proper functioning of Ceramic Coatings in ultra high temperature applications.
- So, there is requirement of Coatings which is having all these attributes and properties which can make the surface of any equipments that are working in ultra-high temperature environment to sustain as long as possible.

2.4 Aims and Objectives

8 unique samples of Ceramic Coatings having different compositions which are having all the required features as discussed above will be tested in thermal cyclic environment**.**

Samples of Ceramic Coatings will be made to go through high temperature environment to observe the life cycle of the Coatings in the terms of the generation and propagation of cracks for the purpose of the suitability and use of ceramic coatings in real practical conditions of ultra high temperature coatings which are frequently used in the aerospace industry, military applications, as well as applications in other extreme heat conditions.

CHAPTER 3

METHODOLOGY

3.1 Experiment Overview

- 8 small sample of alumina based Ceramic coatings will be taken and it will be heated at very high temperatures i.e., maximum temperature of 1100℃ , and then it is allowed to cool by passing it through the passage of high velocity flow of cool air (Quenching) and this process will be repeated continuously in a cyclic manner.

  The sample coatings for accelerated thermal testing in this program were selected as:
  First Set:
  AZ
  AZC
  A8Y
  A8YC
  Second Set:
  A3Y
  A3YC
  A8YCe
  Al_2O_3



Fig. 2.5 Ceramic Coatings sample images

Bi-layer (Top-A8YCe, middle-A3YC) micro Al_2O_3. Following were the parameters which were kept for the Plasma Spraying of the Coatings at IIT Patna:

For Micro Alumina, Current: 708 Amperes, Voltage: 57.2 Volts, Power: 35-40 kW, Argon: 80-100 Scfh, Hydrogen: 12-15 Scfh, Feedrate: 12-15 gm/min, Stand off: 80-100 mm, Carrier flow: 10 Scfh, Supply Pressure: 60 psi

For (AZ, AZC, A3Y, A3YC, A8Y, A8YC, A8YCe) Current: 708 Amperes, Voltage: 57.2 Volts, Power: 35-40 kW, Argon: 80-100 Scfh, Hydrogen: 12-15 Scfh, Feedrate: 12-15 gm/min, Stand off: 80-100 mm, Carrier flow: 10 Scfh, Supply Pressure: 60 psi.

- This type of Thermal shocks and the accelerated thermal cyclic tests corresponds to the actual start up and shut conditions.

- During the process of the continuous cyclic heating and cooling, residual stress will be generated which leads to generation of segmentation cracks which are visible along the cross - section immediately and can result in premature failure of the coating by spallation and cracking.

- The fracture properties of thermal barrier coatings are dependent on the state of residual stress in the coating and are very critical to with stand the severe thermal fatigue conditions.

Finally, the crack formation and its propagation is continuously monitored to determine the residual stress in the alumina based TBCC by using "ImageJ" Java based image process program/software. So, in this way, we can assess the residual life of Alumina based Thermal Barrier Ceramic Coatings (TBCCs).

Technologies and Equipments Used
1. Optical Microscope
2. Oxy Acetylene Gas Flame
3. Image J software
4. Infrared Thermometer

3.2 Composition and Nomenclature

Composition	Nomenclature
μ-Al_2O_3+n-Al_2O_3 + 20 wt% ZrO_2	AZ
μ-Al_2O_3+n-$Al_2O_3Al_2O_3$+20 wt% ZrO_2+4 wt% CNT	AZC
μ-Al_2O_3+n-Al_2O_3+ 20 wt% 3YSZ	A3Y
μ-Al_2O_3+n-Al_2O_3+20 wt% 3YSZ+4 wt% CNT	A3YC
μ-Al_2O_3+n-Al_2O_3 + 20 wt% 8YSZ	A8Y
μ-Al_2O_3+n-Al_2O_3+20 wt% 8YSZ+4 wt% CNT	A8YC
μ-Al_2O_3+n-Al_2O_3+20 wt% 8YSZ + 20 wt% CeO_2	A8YCe

Fig. 2.6 Composition and Nomenclature of Samples

3.3 Properties of Composition Materials

1. **Alumina (Al_2O_3)**
   i. High corrosion-resistance
   ii. High hardness
   iii. Not oxygen Transparent

2. **YSZ (Yttria-stabilized zirconia)**
   i. High thermal expansion coefficient
   ii. Low thermal conductivity
   iii. High thermal shock resistance

3. **CeO_2 (Cerium oxide)**
   i. Higher thermal expansion coefficient
   ii. Lower thermal conductivity

4. **ZrO_2 (Zirconium dioxide)**
   i. Low thermal conductivity
   ii. Very good wear resistance
   iii. Very high fracture toughness
   iv. Resistance to corrosion

3.4 Experimental Procedure

1. After finally building the setup and successfully completing the trials, experimental tests is started with the Ceramic Coatings.
2. For testing, each coating sample was positioned in one of the grooves of graphite block one by one or it can be said that Ceramic Coating sample is then fixed in one of the grooves of the block which is set vertically and is set at 2 inches from the Oxy-Acetylene flame.
3. A purpose-built oxyacetylene torch test rig was used to perform high-temperature testing.
4. Then Oxy-Acetylene flame is made to start and by using a pressure regulator, the flame is set at constant heat emission.
5. Now, this constant flame is applied to the slice (small cut out from the upper portion of the block about 1.5 cm thick) which act as a shield to avoid the direct exposure of the flame on the samples so the direct heat starting from low temperature is prevented on the Ceramic Coatings sample and the heating process starts and the temperature of the cut-out part starts increasing.
6. The temperature variation concerning time is measured by using a Pyrometer (Non-Contact Infrared Thermometer) and it is set at a constant distance from the workpiece by using its stand.
7. When the temperature of the cut out rises to 1100ºC immediately it is taken away and then the flame is directly made to strike the Ceramic Coatings sample at such a high temperature.
8. Now when the temperature of the Coatings sample is made to rise again to 1100°C, and then further it is heated for 60s to rise the temperature beyond 1100°C.
9. After 60s flame is cut off and the Ceramic Coatings are allowed to cool naturally.
10. We are required to note the time when temperature of Ceramic Coatings sample returns again at 1100°C.
11. Now further it is allowed to cool naturally.
12. The temperature variation concerning time is measured by using a Pyrometer (Non-Contact Infrared Thermometer) and it is set at a constant distance from the workpiece by using its stand.
13. The surface microstructure of each Ceramic Coatings sample after oxyacetylene torch testing was characterized using a field emission optical microscopy.

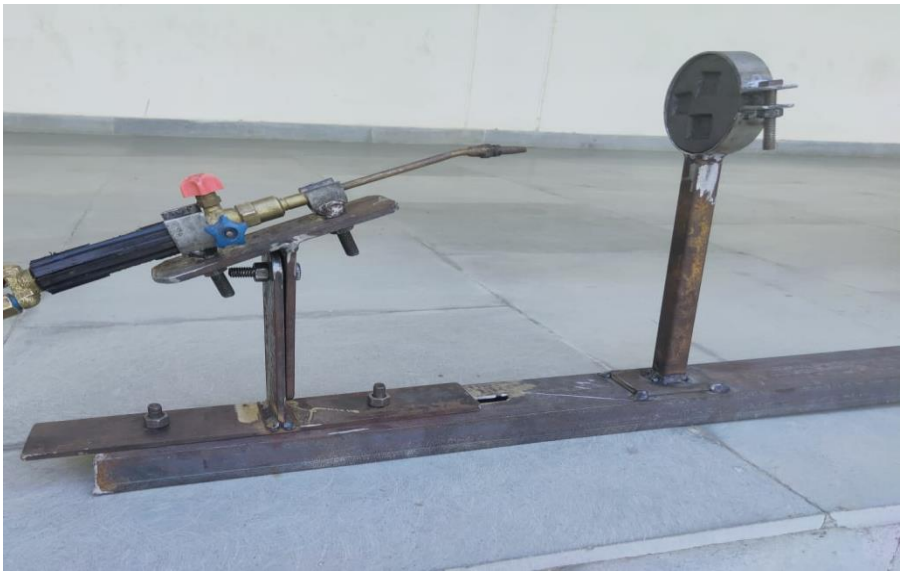


Fig. 2.8.1 Experimental setup image



Fig. 2.8.2 Grooves in Graphite Block

CHAPTER 4

RESULT AND ANALYSIS

Even though Alumina based Ceramic Coatings has good thermal stability, a low thermal conductivity, a high coefficient of thermal expansion (CTE) in combination with a high fracture toughness amongst all the potential materials for UHTC applications, the results of the oxyacetylene torch testing proved its inability to some extent to fulfill the requirements for such applications.

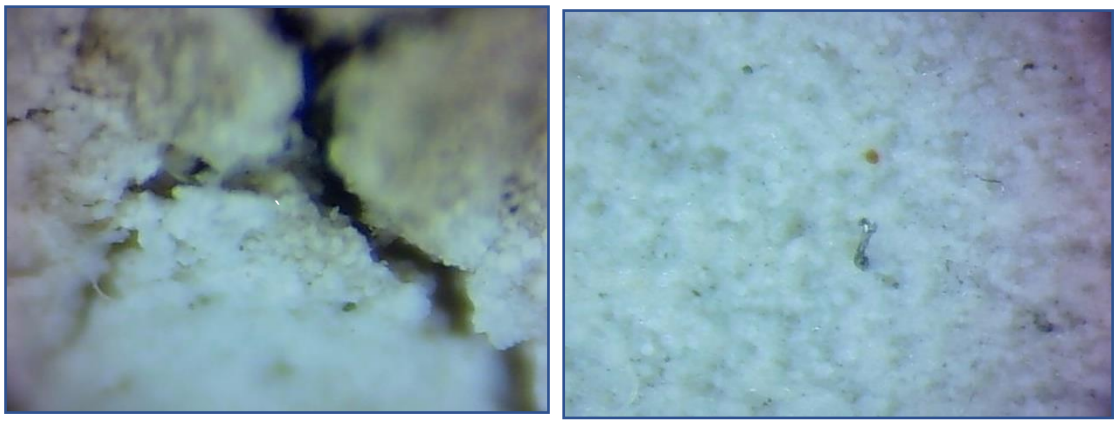
TBCC which includes samples of AZC, AZ, A8YCe, Al_2O_3, A8YC, A8Y, A3YC, Ceramic coatings was unable to withstand the high thermal shocks experienced during the test, leading to severe cracking, rendering these materials unsuitable for applications requiring ultra-high temperature (>1100◦C) resistance.

However, TBCC which includes samples of A3Y was able to withstand the high thermal shocks experienced during the test, leading to its suitability for UHTC applications.

4.1 Images of the samples before and after exposure to the Oxy-Acetylene flame

All images are taken by optical trinocular microscope (Weswox Optik SZM- 180)
All the images are magnified at 120x. (after heating images are on the left hand side and before heating images are on the right hand side).

1. AZ



Cross Sectional Image of AZ

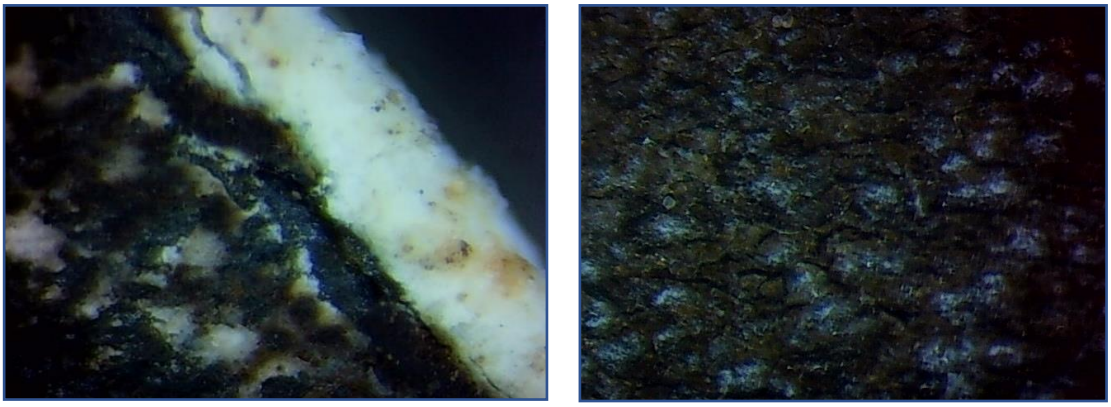
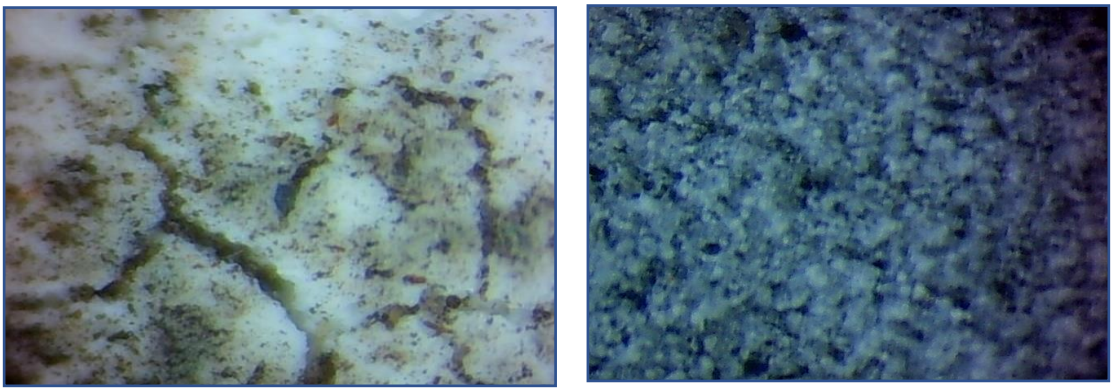


Fig. 3.1 Images of AZ before and after exposure to the Oxy-Acetylene flame

2. AZC



Cross sectional Image of AZC

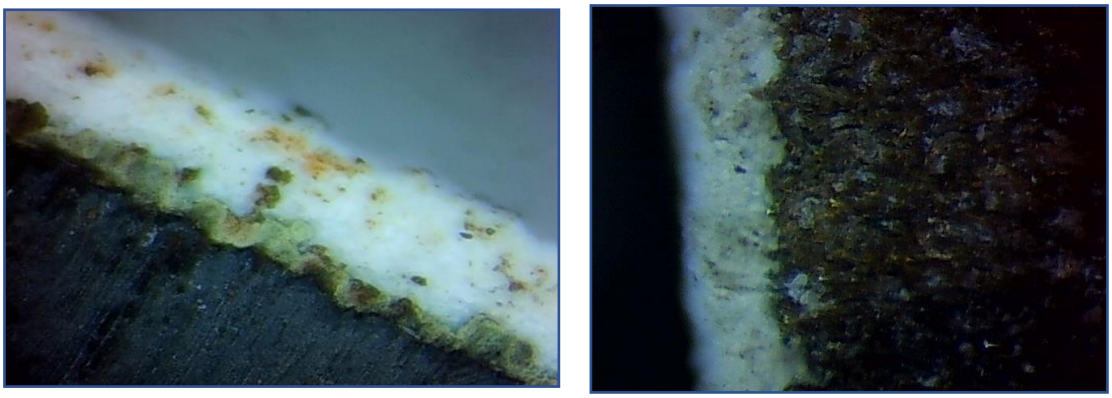
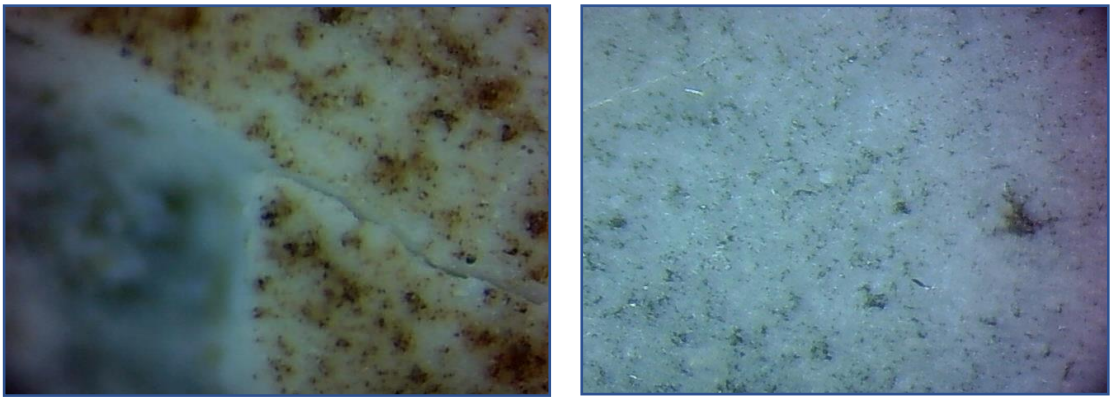


Fig. 3.2 Images of AZC before and after exposure to the Oxy-Acetylene flame

3. Al_2O_3



Cross Sectional Images of Al_2O_3

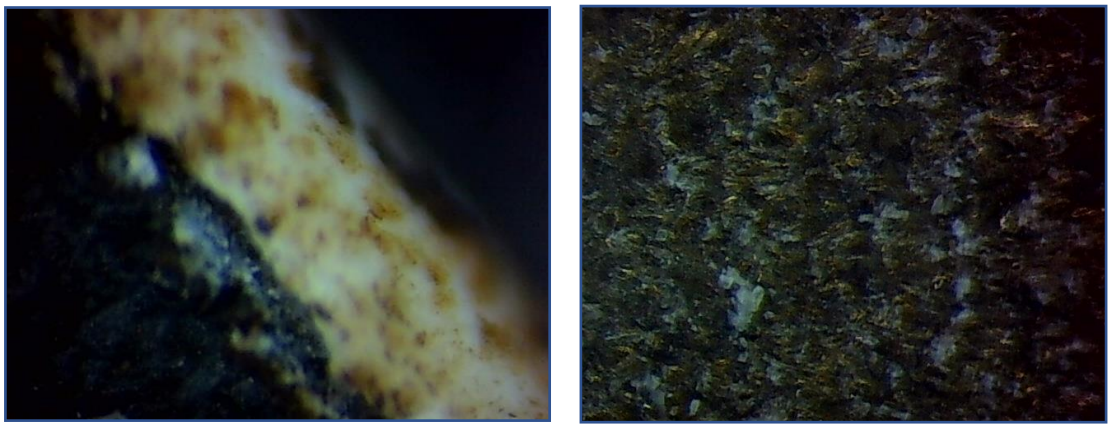
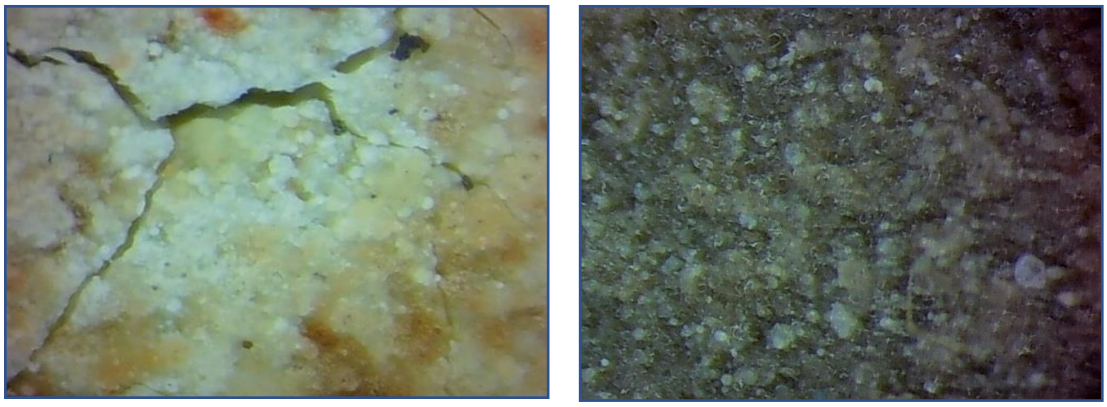


Fig. 3.3 Images of Al_2O_3 before and after exposure to the Oxy-Acetylene flame

4. A8YCe



Cross Sectional images of A8YCe

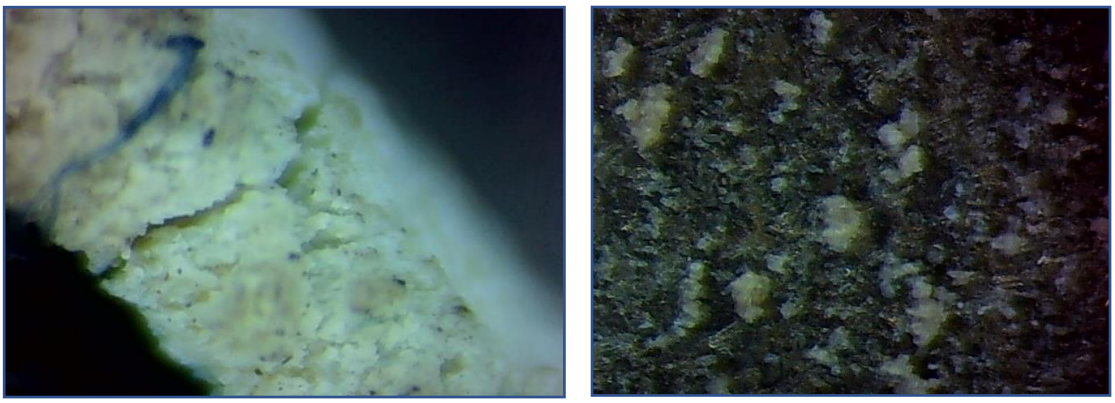
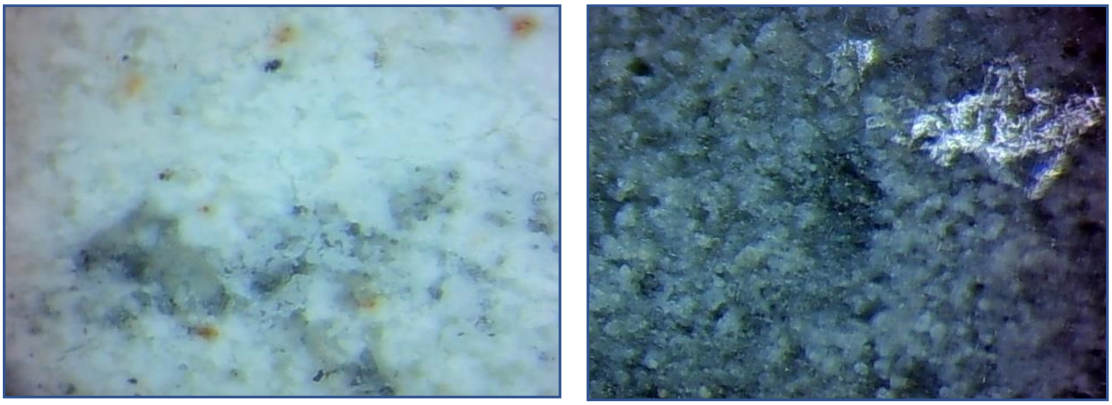


Fig. 3.4 Images of A8YCe before and after exposure to the Oxy-Acetylene flame

5. A8YC



Cross Sectional images of A8YC

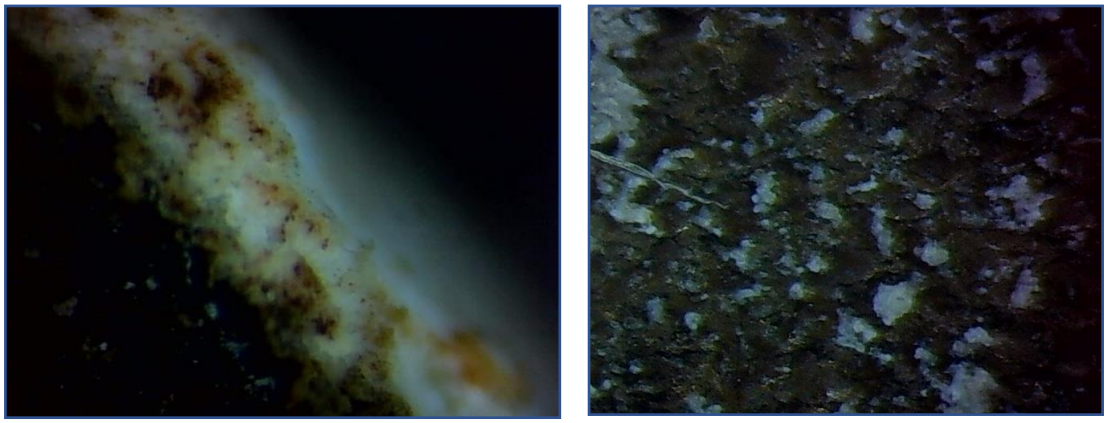
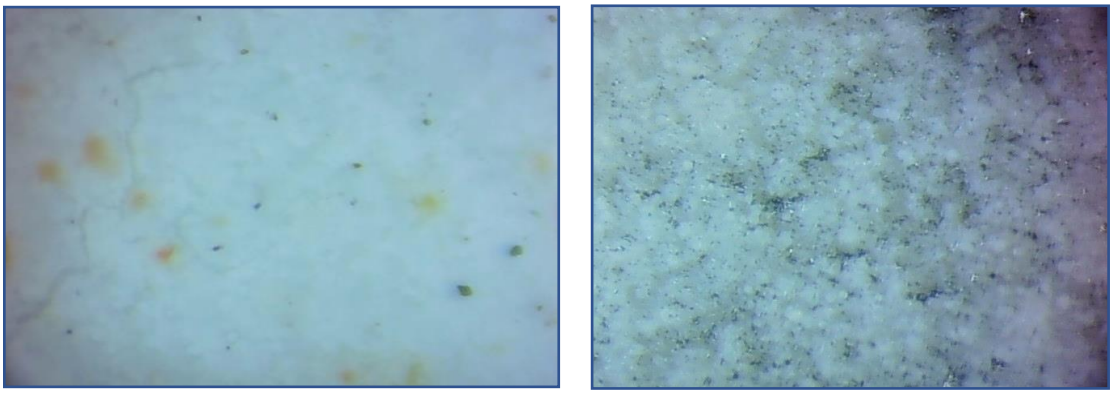


Fig. 3.5 Images of A8YC before and after exposure to the Oxy- Acetylene flame

6. A8Y



Cross Sectional Image of A8Y

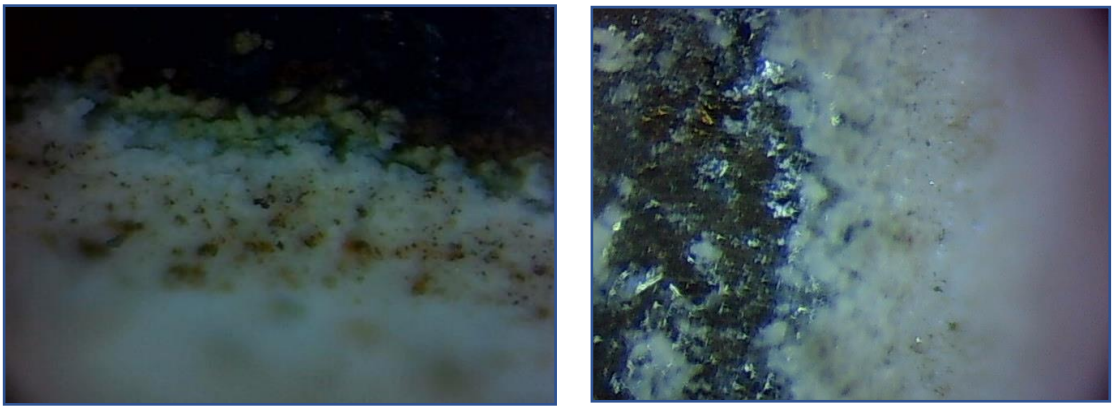
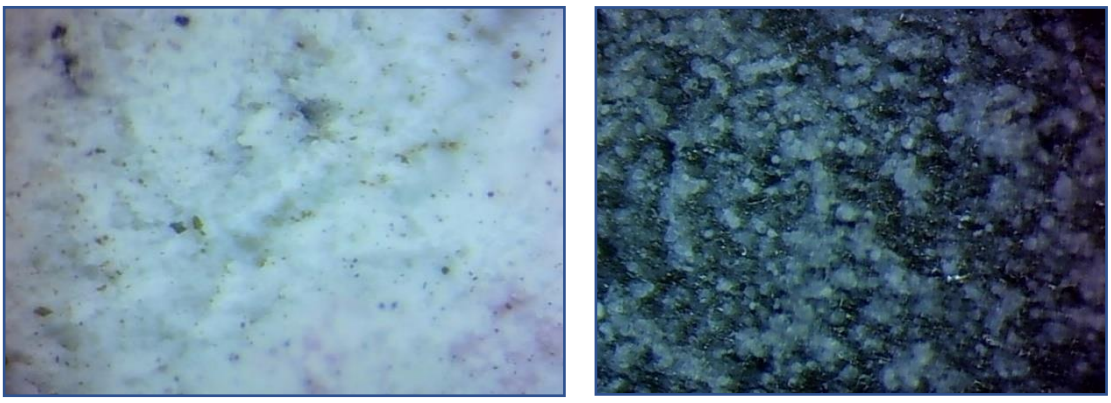


Fig. 3.6 Images of A8Y before and after exposure to the Oxy-Acetylene flame

7. A3YC



Cross Sectional images of A3YC

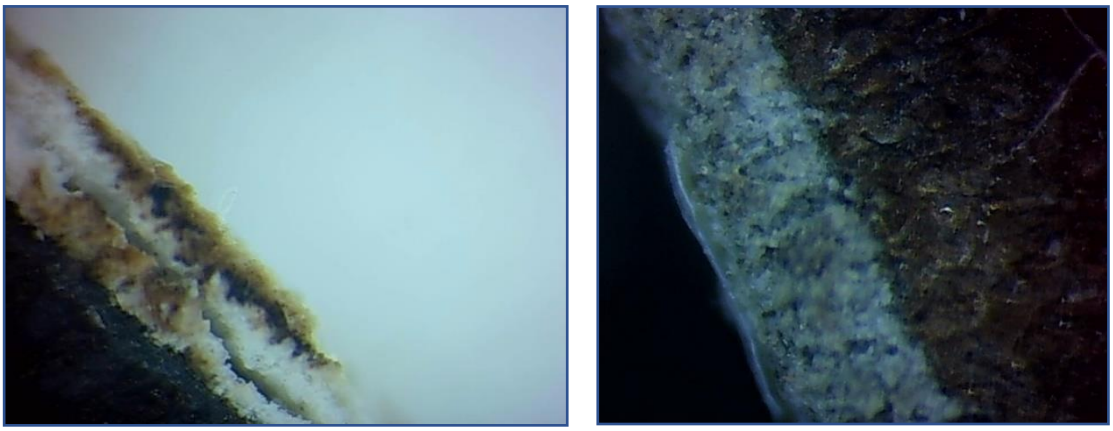
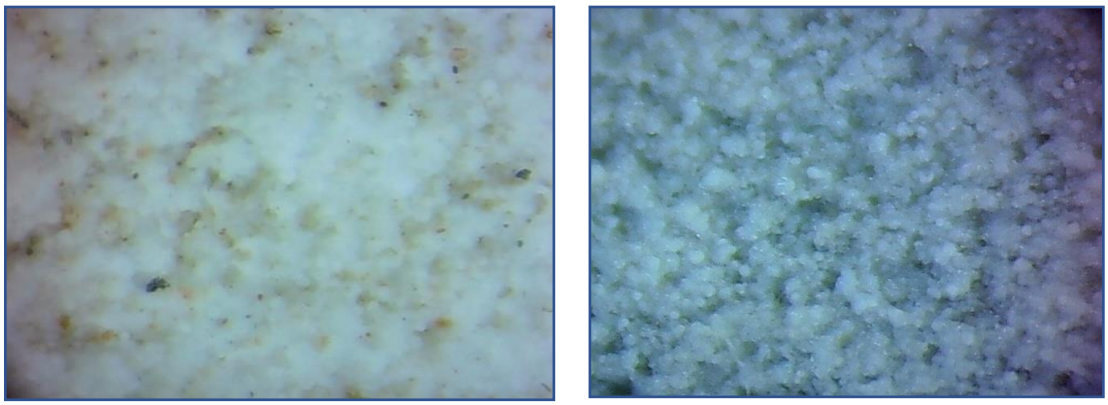


Fig. 3.7 Images of A3YC before and after exposure to the Oxy- Acetylene flame

8. A3Y



Cross Sectional image of A3Y

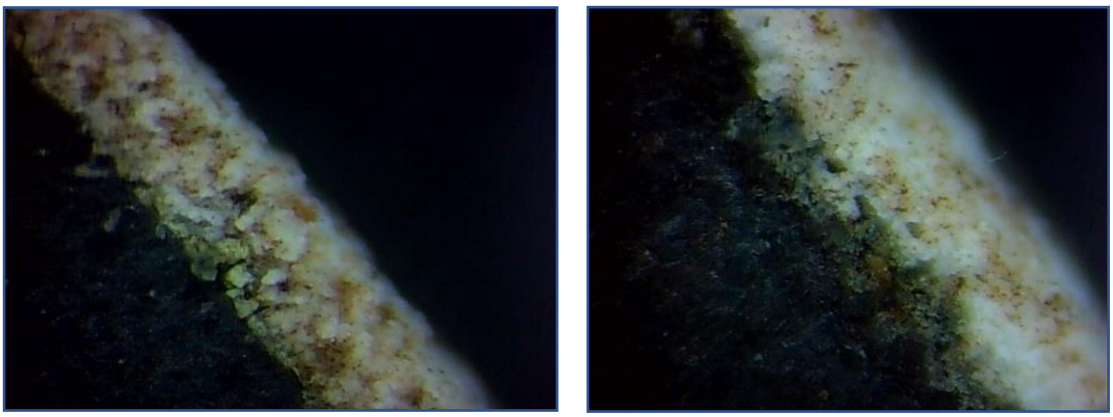
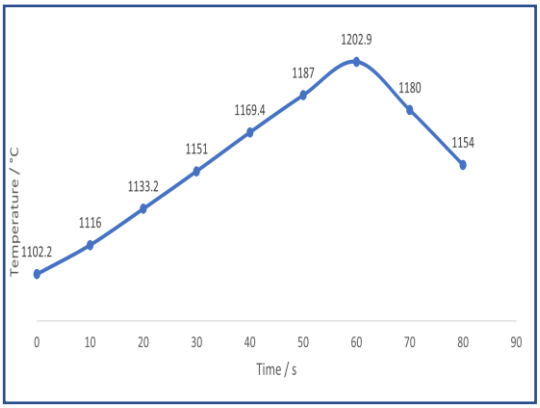


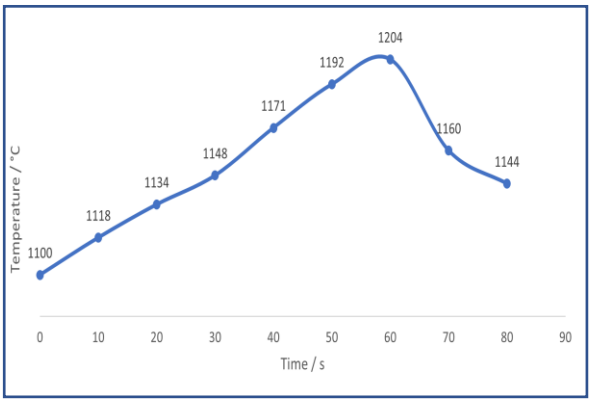
Fig. 3.8 Images of A3Y before and after exposure to the Oxy- Acetylene flame

4.2 Graph representing the Temperature profile with respect to time of all the samples of Ceramic Coatings.

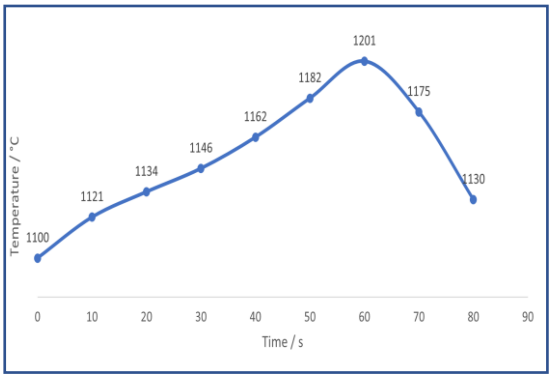
1. AZC



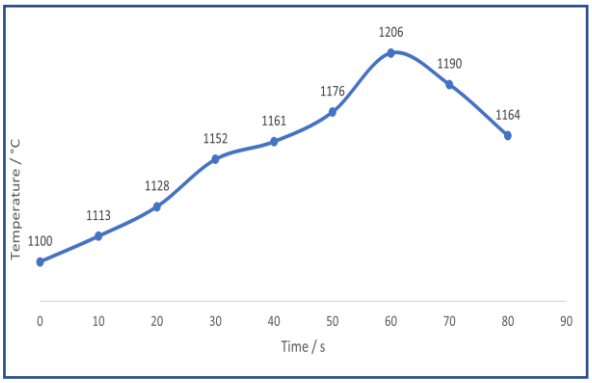
2. A8Y



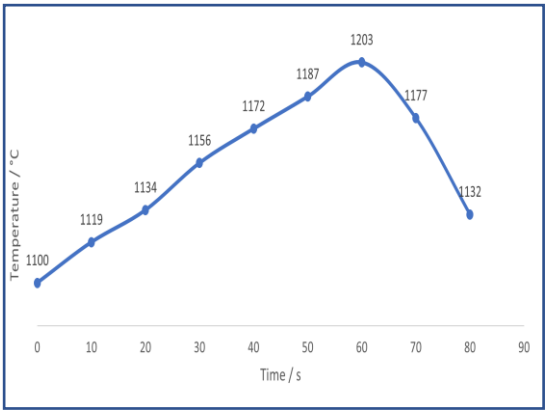
3. A8YCe



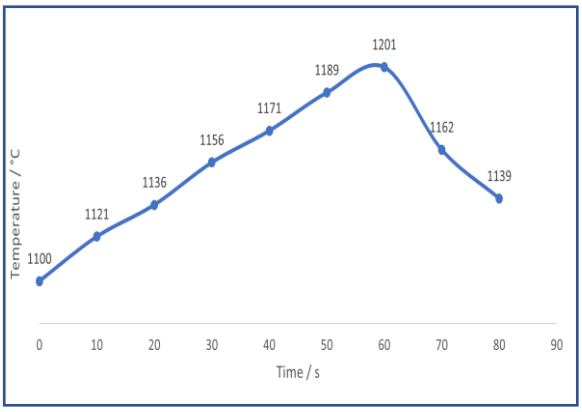
4. A3Y



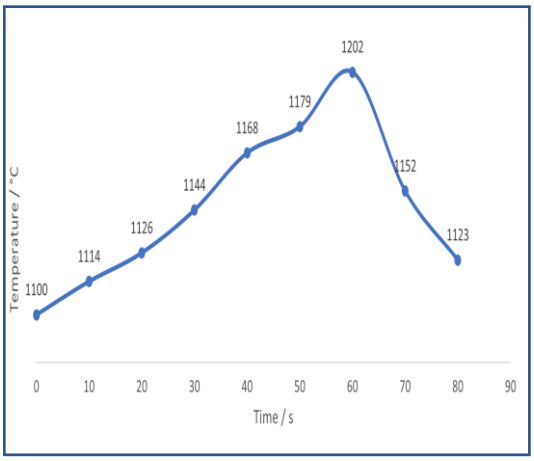
5. A3YC



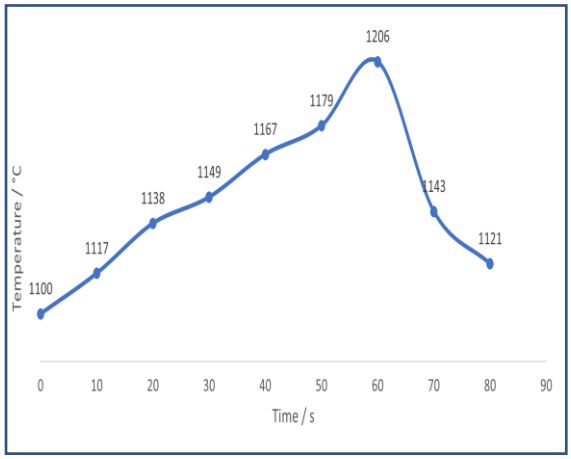
6. AZ



7. A8YC



8. Al_2O_3



CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

The graphical representation of the samples for the crack formation after testing through one thermal cycle is shown in below-

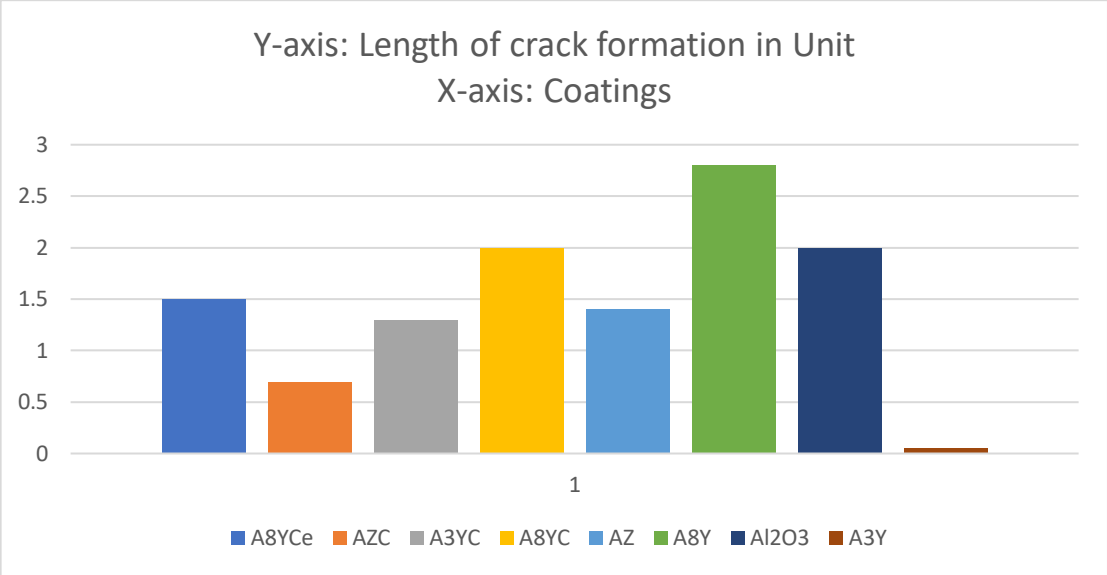


Fig. 4.1 Bar Graph for Crack Formation

The detailed microstructural analysis of the sample after testing revealed several interesting features.

On cross-sectional examination, the cracks were found to propagate into the bulk.

During the test, there was rapid heating and severe thermal gradients. As the samples tried to expand, the Graphite grooves support resulted in stresses being generated that resulted in crack formation.

The cracks originated from the Centre of the sample, where the oxyacetylene-flame was focused, and propagated to the outer points where it was held by the grooves of graphite.

CHAPTER 6

FUTURE SCOPE

Our Ceramic Coatings sample after thermal cyclic tests will further proceed through Scanning Electron Microscope (SEM) imaging for further advancements and research on Alumina based Ceramic Coatings for Ultra high temperature coatings applications (UHTC).

CHAPTER 7

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